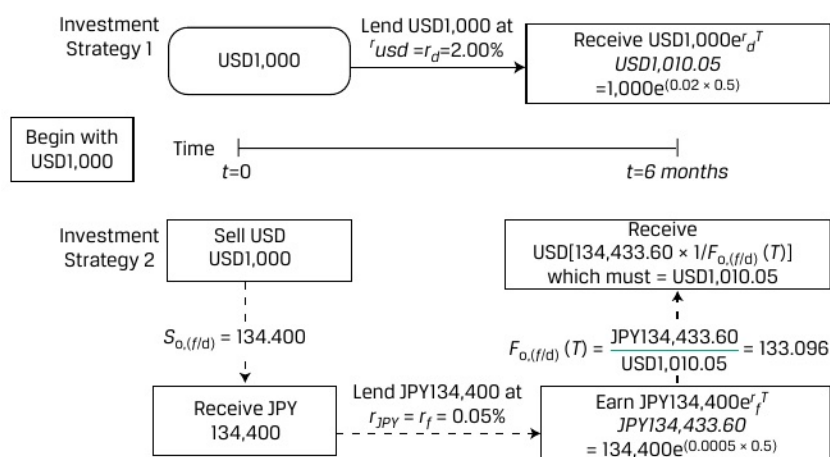


**Exhibit 12: No-Arbitrage Condition in the Foreign Exchange Market****EXAMPLE 14****Foreign Exchange Forward Rates in a Changing Interest Rate Environment**

Central banks responded differently to the sharp rise in inflation during 2022. For example, while the Fed raised rates by 75 bps to 1.75 percent in mid-June, the Bank of England opted for a more gradual approach, raising its benchmark rate by just 25 bps to 1.25 percent. Exhibit 13 compares one-year US Treasury strip rates ( $r_{USD}$ ) from the prior example to UK gilt strip rates ( $r_{GBP}$ ) over the same time frame:

**Exhibit 13: Comparison of US Treasury Strip Rates**

| Date         | $r_{USD}$ | $r_{GBP}$ |
|--------------|-----------|-----------|
| 31 May 2022  | 2.012%    | 1.291%    |
| 15 June 2022 | 2.667%    | 1.562%    |

1. If we assume the USD/GBP spot price of 1.2602 (or USD1.2602 per GBP1.00) from 31 May remains constant, how does the change in risk-free US dollars versus British pounds rates affect the one-year USD/GBP forward rate?

**Solution:**

The no-arbitrage USD/GBP forward rate increases from 1.2693 to 1.2742. Assume an investor has GBP1,000 to invest for one year in a British-pound or US-dollar risk-free discount bond. The US dollars needed to purchase GBP1 in one year must have a spot price equal to the discounted future price.

As of 31 May: